ATU-Tutor: A RAG-Based Socratic Tutoring System with Agentic AI Adaptive Assessment

Md Shaba Sayeed
Engineering and Computing Sciences
Arkansas Tech University
Russellville, Arkansas, USA
msayeed@atu.edu

Shaira Wajiha
Engineering and Computing Sciences
Arkansas Tech University
Russellville, Arkansas, USA
swajiha@atu.edu

Robin Ghosh
Engineering and Computing Sciences
Arkansas Tech University
Russellville, Arkansas, USA
rghosh@atu.edu

Md Abdus Salam Siddique
Engineering and Computing Sciences
Arkansas Tech University
Russellville, Arkansas, USA
msiddique@atu.edu

Shruti Bhandari
Engineering and Computing Sciences
Arkansas Tech University
Russellville, Arkansas, USA
sbhandari3@atu.edu

2026 IEEE World AI IoT Congress (AIIoT) | 979-8-3315-4568-0/26/$31.00 ©2026 IEEE | DOI: 10.1109/AIIoT68874.2026.11569566

***Abstract*—Large language models (LLMs) have great potential for educational tutoring, but hallucinations, poor curriculum alignment, and limited access to domain specific knowledge limit their application in academic settings. Retrieval augmented generation (RAG) is one way to overcome these shortcomings by anchoring model outputs to external sources. Still, most educational RAG systems are limited to single turn question answering and offer little pedagogical assistance. The paper will provide the design and assessment of the RAG enhanced LLM chatbot in Socratic tutoring and adaptive quiz generation at Arkansas Tech University (ATU). The system employs a hybrid retrieval pipeline with FAISS, BM25, and cross encoder re ranking on a knowledge base of 2,328 chunks from 17 open educational resources textbooks, as well as a retriever fine tuned on 929 domain specific query passage pairs. The Qwen3 8B model supports both tutoring and assessment. In tutoring mode, the system generates grounded explanations and Socratic follow up questions; in assessment mode, it generates curriculum aligned multiple choice and short answer questions with immediate feedback. The assessment of 14 questions demonstrated strong results in relevance, conciseness, contextual grounding, and resistance to hallucinations, with overall heuristic and LLM scores of 4.00/5. Lower scores in empathy and Socratic scaffolding indicate areas for future improvement. The findings show the potential of retrieval based tutoring and self adaptive assessment in a university specific curriculum.**

***Index Terms*—Retrieval Augmented Generation (RAG), Large Language Models (LLM), Socratic Tutoring, Agentic AI, Adaptive Quiz Generation, Intelligent Tutoring Systems, Higher Education, FAISS, BM25, Cross Encoder Reranking, Prompt Engineering, Open Educational Resources (OER), Curriculum Aligned Assessment, Natural Language Processing (NLP), Educational Technology**

I. Introduction

Large language models (LLMs) have become a growing focus in education due to their ability to generate natural language responses, interactively answer questions, and provide individualized learning experiences [1], [2], [3]. In post secondary education, these abilities enable scalable tutoring mechanisms that help students access on demand academic assistance outside the classroom, particularly in environments where individual teacher care is limited [1], [2]. More recent research has demonstrated growing concern about conversational AI systems capable of explaining concepts, answering follow up questions, and supporting inquiry based dialogue in realistic academic settings [2]. This tendency is a symptom of a broader shift in the model of content delivery, from a one way approach to incorporating more interactive, student centered education with the help of AI [4]. Despite this potential, general purpose LLMs are inadequate as academic tutors and cannot be trusted to be reliable and safe when not grounded or maintained under pedagogical control [3]. They can include hallucinated, misleading, or factually inaccurate information in their output, particularly when accuracy, credibility, and curriculum coherence are needed [4]. Moreover, these models typically rely solely on static pretraining data. They thus may not fully capture the institution specific course material, the goals of the current instruction, or the terminology used in a given discipline [1]. Consequently, efficient academic tutoring systems should not simply produce plausible answers but also provide systematic, credible, and pedagogically significant assistance that guides students in accumulating knowledge step by step [5], [6], [7]. To overcome these constraints, RAG offers a pragmatic approach by leveraging model outputs based on curated external sources of knowledge, including textbooks, teaching resources, and academic documents [1]. The RAG based system retrieves passages of a document collection that are relevant to the query but are not based on parametric knowledge, and makes these passages available as contextual evidence as the response is generated. This design helps increase grounding in fact, reduces the possibility of hallucination, and enables the inclusion of domain specific information without retraining a whole language model [3]. These properties are especially valuable in education, as they

979-8-3315-4568-0/26/$31.00 ©2026 IEEE

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:58:37 UTC from IEEE Xplore. Restrictions apply.

allow a tutor's response to align with authoritative learning materials rather than with Web scale priors that are more widely disseminated and uncontrolled [3]. Factual grounding, however, is not in itself enough to make a good tutor. Even most of the AI and RAG based systems currently used in education remain primarily question answer systems, searching for information and providing explicit answers without truly supporting the pedagogical interaction required to drive real learning [4]. Incorrect tutoring involves systematic expositions, directed inquiries, comments on incomplete knowledge, and opportunities for self evaluation [6]. Specifically, the Socratic type dialogue is pedagogically useful in that it prompts students to express their arguments, challenge their assumptions, and actively engage with the content rather than passively accept the answer [7]. Based on these considerations, one can notice a research gap: educational systems that integrate grounded retrieval into dialogic tutoring and adaptive assessment within a single framework are required [4]. The gap in quiz based learning is very important. Assessment is not only applicable to grading but also to strengthening comprehension, uncovering misunderstandings, and enabling learners to check their own development [5]. Multiple choice and short answer questions, when aligned with instructional material and followed by immediate feedback, may facilitate retrieval practice, reinforce understanding, and enhance self regulated learning [7]. However, in most AI learning systems, tutoring and assessment remain distinct, even though they are inherently complementary in actual educational practice [5], [4]. The ability to explain content and generate grounded and curriculum relevant quizzes, therefore, provides students with a comprehensive and educationally significant learning workflow through a tutoring system [8]. To mitigate these shortcomings, this paper introduces a RAG enhanced LLM chatbot that can serve ATU students and facilitate Socratic tutoring and adaptive quiz generation. The system utilizes a knowledge based retrieval pipeline consisting of dense semantic retrieval, sparse lexical retrieval, and neural reranking to enhance evidence selection and domain alignment, based on a body of knowledge comprising 17 open educational resource textbooks [9]. The assessment and the tutoring module retrieve content relevant to producing grounded explanations and scaffolding questions to promote active reasoning [10], [11] and the assessment module produces multiple choice and short answer quizzes grounded in feedback and aligned with the curriculum [12], [13]. The proposed system is designed exclusively for tutoring and formative assessment and does not require the collection of sensitive personal information. When deployed locally or in a controlled environment, user queries do not have to be exposed to external internet services and are thus less prone to privacy and data leakage. The proposed system merges grounded retrieval, conversational tutoring, Socratic scaffolding, and adaptive assessment within a university congruent architecture as opposed to generic educational chatbots [14] , [12]. It is a design that goes beyond mere academic question answering to a better grounded, more pedagogically organized form of AI assisted learning in higher education [15], [16].

II. Related Work

Recent educational tutoring systems are progressively integrating retrieval augmented generation (RAG) with large language models (LLM) to enhance factual grounds, instructional relevance, and learner adaptation. LPITutor reflects this trend, combining LLMs, RAG, and prompt engineering to generate tailored tutoring responses based on learner proficiency and question difficulty [17]. Likewise, RoboTuB demonstrates that domain specific learning can be facilitated through retrieval grounded tutoring, which creates multi step STEM problems using structured explanations, simulations, and code examples [18]. Nevertheless, such systems primarily focus on individualized support and responsive material presentation, at the expense of direct Socratic conversation or built in formative evaluation. Another parallel line of research concerns Socratic and dialogic tutoring. A notebookLM is a Socratic physics tutor in which retrieval grounded, source based interaction is employed to provide students with question based learning, demonstrating that RAG can facilitate more traceable and pedagogically organised tutoring [19]. The Socratic AI for clinical case based learning also uses an adaptive strategy similar to the LLM based tutoring approach, based on guided reasoning, real time feedback, and systematic assessment of learning strengths and weaknesses [20]. The studies are closely related to our system's tutor system, but they are usually area specific, lack shared context, broad academic corpus grounding, and a separate assessment generation component. Regarding the assessment side, it has previously been demonstrated that RAG and LLM stimulation are applicable for producing adaptive quizzes with high topical and contextual relevance [21]. Other researchers also show that course aligned retrieval and domain adaptation can enhance the quality of tutoring in curriculum specific environments [22]. Though this has been done, it can still be said that the current work remains disjointed among the individualized RAG tutoring, Socratic instructional dialogue, and the generation of quizzes, which are mostly distinct. On the other hand, our system combines a hybrid retrieval pipeline for course relevant textbooks, a retrieval grounded Socratic tutoring module, and an agentic quiz generation workflow into a single university centric architecture for guided learning and formative assessment.

III. Methodology

A. System Overview

This paper describes the design and development of a Retrieval Augmented Generation (RAG) system based on an LLM and an agentic quiz generation workflow to aid Socratic tutoring and formative assessment of students at Arkansas Tech University (ATU). The system architecture comprises of five major components: (1) a knowledge base of domain specific knowledge of open educational resource textbooks applicable to ATU coursework, (2) a hybrid retriever based on dense vector retrieval, sparse lexical retrieval, and neural reranking, (3) a fine tuned dense retriever, tailored to the

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:58:37 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF EXISTING RAG-BASED TUTORING AND QUIZ GENERATION APPROACHES

Reference	Proposed System	Finding	Limitation
Liu et al., 2025	LPITutor	RAG and prompt engineering improve the quality of personalized tutoring, including accuracy, clarity, and coherence.	Does not explicitly support Socratic tutoring or agentic quiz generation.
Pandey et al., 2025	RoboTuB	Retrieval-grounded tutoring supports multi-step STEM learning and improves learner outcomes.	Focused on STEM MOOCs, not a broad university chatbot with Socratic dialogue and quiz workflow.
Tufino, 2025	NotebookLM as a Socratic Physics Tutor	RAG can support traceable Socratic tutoring grounded in source documents.	Domain-specific and does not include a separate quiz-generation module.
Golchini et al., 2025	Socratic AI	Adaptive Socratic tutoring scaffolds learners' reasoning through feedback.	Specialized for clinical case-based learning rather than general academic tutoring.
Gopi et al., 2024	Adaptive Quiz Generation with RAG	RAG enables context-grounded adaptive quiz generation with relevant explanations.	Quiz-centric and not unified with a Socratic tutoring chatbot.
Karatepe et al., 2026	RAG + Domain Adaptation for ITS	Course-aligned retrieval and adaptation improve the relevance and coherence of tutoring.	Framed more as tutoring QA than a Socratic tutor with agentic formative assessment.

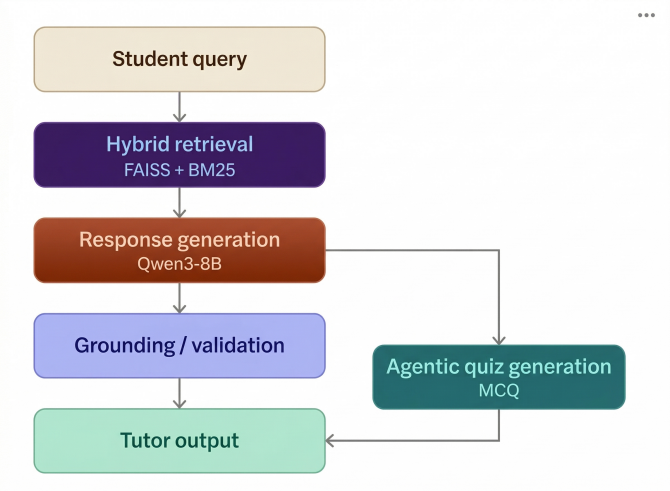


Fig. 1. Workflow of the ATU-Tutor System.

academic corpus, (4) an LLM based Socratic tutoring module and (5) an agentic quiz generation module to perform curriculum aligned formative assessment. It was run on a Kaggle Notebook system with a chat interface built with Gradio that provides real time, retrieval grounded tutoring.

B. Knowledge Base Construction

The knowledge base was constructed from 17 Open Educational Resources (OER) textbooks retrieved from Arkansas Tech University's Online Research Commons across a variety of academic subjects. All the books were gathered as PDFs and processed page by page with PyPDF. Removal of page numbers, text page breaks, and page whitespace; normalization of Unicode characters; and repair of hyphenation were all done to extract text and cleanse it. The cleaned corpus was then segmented using a sliding window chunking method with chunks of about 900 tokens and a 140 token overlap; very short chunks were discarded. This fixed length overlapping design was used as a practical baseline for long document retrieval, as has been done before and in line with previous studies demonstrating that chunking quality and overlap affect long document retrieval performance [23]. A chunk was assigned a SHA 1 identifier and stored with metadata, including title, source path, page range, and an approximate token count, to support reproducible provenance and indexing [24] [25]. The pipeline created 2,328 indexed chunks, stored in JSONL format for downstream access. The system specific information below corresponds to the uploaded methodology description.

C. Hybrid Retrieval Pipeline

A hybrid retrieval pipeline has been designed to maximize the relevance and coverage of the context provided to the LLM. The pipeline combines three retrieval mechanisms that are complementary, each more effective in its own right: dense semantic retrieval, sparse lexical retrieval, and neural cross encoder re ranking [26]–[28]. Dense Vector Retrieval: Each chunk in the knowledge base was represented as a 384 dimensional dense vector using the Sentence Transformer model, all MiniLM L6 v2. The embeddings were L2 normalized and indexed with a FAISS IndexFlatIP inner product index to perform efficient similarity search on the corpus [29]. Sparse Lexical Retrieval: In parallel, a BM25Okapi index was built over the tokenized corpus. Tokenization included lowercasing, removing invalid characters, and removing tokens shorter than 3 characters. This component captures exact lexical matches for domain specific terminology that may not be fully preserved in dense embedding space [26], [28]. Score Fusion: For each incoming query, dense retrieval and BM25 retrieval were performed separately, and each returned the top 50 candidate chunks. The two ranked lists were fused (weighted linear combination with weights of 0.6 for dense retrieval scores and 0.4 for BM 25 scores).These weights were chosen as an intuitive heuristic to give greater weight to semantic similarity and keep lexical matching for course-specific terms. A complete ablation study of different dense and sparse retrieval weights remains to be done. A soft boilerplate filter was used for the front matter (license page, table of contents, citation packed sections, etc.), and a minimum of 12 candidates were retained to avoid retrieval failure in edge cases [28]. Cross Encoder Reranking: The fused candidate set was reranked using the cross encoder/ms marco MiniLM L6 v2 model, which jointly encodes each query passage pair to obtain fine grained relevance scores. The five best passages/propositions re ranked were used as the last retrieval context for the generation module [27], [30]. This three stage retrieve, fuse,

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:58:37 UTC from IEEE Xplore. Restrictions apply.

and re rank design was intended to balance recall, precision, and computational efficiency [27].

D. Retriever Fine Tuning

To adapt the dense retriever to the domain specific characteristics of the ATU textbook corpus, the base Sentence Transformer model was fine tuned on a custom query passage dataset. Training pairs were constructed using three complementary strategies: (1) 95 hand crafted domain questions paired with BM25_retrieved chunks from the 17 textbooks, (2) synthetic queries generated from 600 randomly sampled chunks by truncating their opening sentences, and (3) cross chunk pairs in the book linking topically related passages within the same textbook. This process produced 929 query passage pairs, which were split into 836 training pairs and 93 validation pairs. Fine-tuning used the MultipleNegativesRankingLoss objective for 3 epochs with a batch size of 32 and a learning rate of 2×10^{-5}, with linear warmup over 10% of the training steps.Mixed precision training was performed on an NVIDIA Tesla T4 GPU, and validation was conducted using an InformationRetrievalEvaluator after each epoch. The best checkpoint was retained and the FAISS index was rebuilt using the embedded data of the updated model so that the subsequent retrieval reflected the domain adapted embedding space [31], [32].

E. Large Language Model Configuration and Inference Controls

The generative aspect of the system is grounded in Qwen3 8B, an 8 billion parameter language model. To allow deployment on a single Nvidia Tesla T4 GPU with 16GB of VRAM, the model was loaded with the 4 bit NormalFloat (NF4) quantization using the BitsAndBytes framework, switching to double quantization and a computation data type of float16. This configuration significantly reduced memory usage while maintaining very good generation quality for tutoring and quiz generation applications [33], [34]. To improve response quality during inference, a lightweight output control pipeline was employed. Internal reasoning traces and meta commentary were removed using pattern based filtering. In addition, to detect outputs with unusually low alphabetic character content, a simple corruption detection heuristic was used to provide a fallback response when needed. Finally, the pad token was tied to the end of sequence token to allow for stable generation behavior [34], [35].

F. Socratic Tutoring Module

The Socratic tutoring module is the main pedagogical interface of the proposed system. It focuses on supporting active learning through very short, retrieval based explanations and a selected follow up question. The LLM was prompted with a structured prompt that elicited a two part response: EXPLAIN, which provides a short answer using only the retrieved notes, and QUESTION, which includes one Socratic follow up question. The prompt precluded outside knowledge, code generation, and vague questioning, and a lack of sufficient evidence triggered a controlled fall back response [36], [37]. To make the system more reliable, topic based query expansion, lightweight affect sensitive routing of confused or frustrated learners, and grounding checks to minimize unsupported generation were employed. In the event of a retrieval failure, the system performed a second expanded search, followed by a fallback message. These mechanisms support concise, grounded, and pedagogically guided tutoring behaviour [6], [38].

G. Agentic Quiz Generation Module

The quiz generation module extends the system from conversational tutoring to formative assessment. It operates as a goal directed, multi step workflow for topic selection, supporting passage retrieval, item generation, output quality checking, and grounding feedback [39], [40]. Selection of Topics: The module includes a pool of quiz topics that can be configured to align with the 17 textbooks. At quiz initialization, topics are either selected by the user or randomly sampled, then divided into multiple choice and short answer fields according to the desired quiz structure [41]. Retrieval Grounded Generation: The module follows the hybrid retrieval pipeline outlined in Section III C to retrieve the top 5 passages for each chosen topic. These notes are then fed to Qwen3 8B with a structured prompt that limits question generation to the retrieved material [42]. For multiple choice questions, the prompt asks the model to generate a single question, four choices, the correct answer letter, and a one sentence explanation, in the form QUESTION, A, B, C, D, ANSWER, and EXPLANATION. In short answer questions, the model should generate one answerable question along with a model response in the form of a question and answer pair, two to three sentences [41], [42]. Output Parsing and Validation: The generated output is purged of reasoning traces, and then regular expressions are used to process it. Questions are validated based on three criteria: all required fields must be present, the question text must be a minimum length, and the explanation or model answer must have sufficient detail. Items that do not pass validation are discarded [43]. Interactive Quiz Delivery: Valid questions are displayed to the student in random order. For the multiple choice questions, the student clicks an answer and receives instant feedback. The grounded explanation is used to reinforce correct responses, whereas the incorrect answers display the correct response and explanation. For short answer items, the model answer is shown to the student after they respond, and the student is invited to self assess their performance on a three point scale [39]. Formative Feedback: A summary report that appears at the end of the quiz session includes multiple choice accuracy, the average self assessment for short answer questions, and a list of topics associated with incorrect answers. This enables the student to identify areas that require further study [43].

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:58:37 UTC from IEEE Xplore. Restrictions apply.

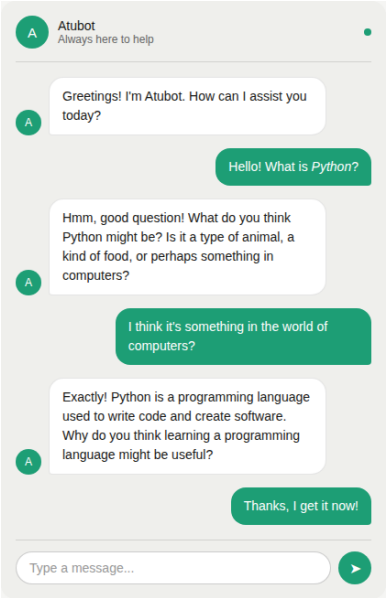


Fig. 2. ATU-Tutor chat interface.

IV. EVALUATION AND RESULTS

A. Evaluation Framework

The suggested system has been tested using a multi dimensional assessment framework developed to quantify the quality of tutoring responses and the pedagogical effectiveness of the Socratic dialogue. The set of 14 questions was prepared to address academic content queries across various subjects, misconception related prompts, emotionally charged student responses, and one boundary testing question that was not within the scope of the primary tutoring. Examples were included: “What is database normalization?” “Why use R instead of Python?” “R is paid software, right?” “I have no idea about this topic”, and “What happened during Hurricane Andrew?” This diversity was meant to provide stress tests for the system based on realistic student interaction patterns, to measure several dimensions of tutoring performance, such as conceptual support, misconception management, affective responsiveness, and safe boundary behavior [5], [6], [44]. The test was an automated, heuristic scoring, rubric directed LLM as judge test. The heuristic measures captured relevance, empathy/instructional tone, conciseness, and context grounding, which are shared dimensions for gauging the quality of educational dialogue and feedback to the learner [6], [45]. The evaluation based on the LLM was used to assess the ability to resist hallucinations, the quality of the Socratic scaffold, and the use of context, using a pedagogically informed rubric. This two trial design aligns with existing research suggesting the use of multi method assessment of educational AI systems, specifically evaluating response quality and assessing the greater teaching value of scaffolded dialogue [37], [46].

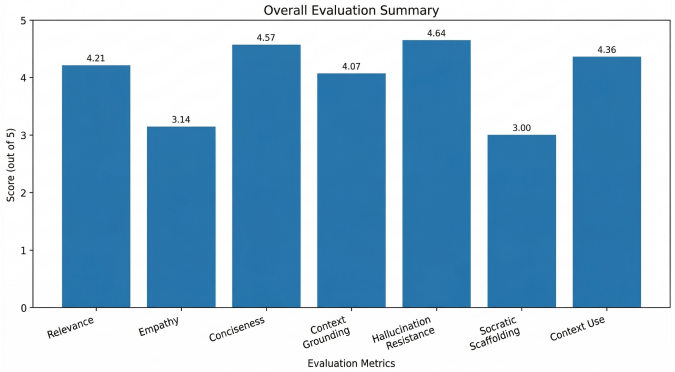


Fig. 3. Overall Evaluation Summary of the ATU-Tutor System.

B. Analysis of Results Matrix

The results Table II indicates that the ATU RAG Socratic Tutor performs best in relevance, conciseness, grounding, and hallucination resistance, while empathy and Socratic scaffolding are weaker. Relevance (4.21/5) suggests that, in general, the system responded to student queries accurately and well domain matched, particularly for well formulated academic queries. The best major measure was hallucination resistance (4.64/5), indicating that the retrieval augmented design effectively limited unsupported or fabricated claims. On the same note, the scores in the context related category, including Context Heuristic (4.07/5), Context Use (4.36/5), indicate that the tutor gave meaningful responses based on the retrieved information, and Conciseness (4.57/5) demonstrates that the tutor gave short and student oriented answers [47], [48]. Empathy (3.14/5), in turn, along with Socratic Scaffolding Quality (3.00/5), was relatively low. Despite the fact that the system regularly maintained a helpful tone and created follow up questions, it did not always go deep enough or be adaptive and pedagogically strategic. Such a trend is consistent with previous studies of tutoring history, which have demonstrated that automated systems tend to be effective at reliably delivering knowledge, albeit with comparable affective sensitivity and adaptive guidance to human expert tutors [48]–[50]. On the whole, the heuristic and LLM based aggregate scores were 4.00/5, indicating high overall performance. These findings imply that the tutor is already satisfactory as an instruction grounded academic support tool. Still, it should be enhanced in the future by incorporating more meaningful scaffolding techniques, misconception related prompts, and more overt, affect sensitive feedback [50]. In a broader sense, current research on educational chatbot evaluation also facilitates the assessment not only of answer quality but also of learner engagement and the perceived value of instruction [5].

C. Per Question Hallucination and Scaffolding Analysis

In Table III, hallucination resistance was very stable across most questions, with almost all responses rated 5/5, indicating that the retrieval augmented design was highly effective at restricting the generation of unsupported facts. The greatest exception was the affective prompt, ”I keep getting this wrong,

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:58:37 UTC from IEEE Xplore. Restrictions apply.

TABLE II
EVALUATION RESULTS

Metric	Score (out of 5)	Percentage
Relevance	4.21	84.2%
Empathy	3.14	62.8%
Conciseness	4.57	91.4%
Context Heuristic	4.07	81.4%
Heuristic Overall	4.00	80.0%
Hallucination Resistance	4.64	92.8%
Socratic Scaffolding	3.00	60.0%
Context Use (LLM)	4.36	87.2%
LLM Overall	4.00	80.0%

TABLE III
PER-QUESTION HALLUCINATION AND SCAFFOLDING SCORES

Question	Hallucination	Scaffolding
What is database normalization?	5/5	3/5
What is 1NF?	5/5	5/5
What is logistic regression?	5/5	3/5
Why use R instead of Python?	5/5	3/5
What is hydrogen bonding?	5/5	3/5
What is material fatigue?	5/5	3/5
What is the Z-transform?	5/5	3/5
What is phonological awareness?	5/5	2/5
What is career advising?	5/5	2/5
What is the digital divide?	5/5	3/5
What happened during Hurricane Andrew?	5/5	3/5
R is paid software right?	5/5	2/5
I have no idea about this topic	5/5	4/5
I keep getting this wrong, I give up	0/5	3/5
Average	**4.64/5**	**3.00/5**

I give up," which got 0/5 since the answer focused on encouragement rather than retrieval grounded content. The quality of the scaffold varied: many questions scored 2/5 or 3/5, while only one reached the maximum of 5/5. This implies that the tutor can create follow up questions, but their pedagogical depth is not always high.

D. Quiz Generation Assessment

The quiz generation agentic module was tested independently of the tutoring element. It has been assessed across various academic subjects using multiple choice and short answer quizzes. The module was able to extract relevant notes and formulate legitimate quiz items based on the scholarly corpus. The practicality of automated quiz building, delivery of formative assessment, and immediate feedback processes in learning institutions can be justified by the previous research on digitally supported and AI enabled assessment [51], [52]. For multiple choice questions, the system generated a question stem, four viable response choices, a correct response, and a brief response using retrieved information. The distractors were intended to reflect probable misconceptions rather than random options, which is in line with the current assessment design paradigm [53], [54]. For short answer questions, the system would generate a question and a brief model answer based on the supporting notes. Response tracking, immediate feedback, and a formative summary indicating the aspects that needed to be reviewed again were also supported by the interactive quiz workflow, which is consistent with the literature on digital formative assessment [55], [56]. These findings suggest that an agentic quiz item can independently convert the retrieved course content into an applicable formative assessment. In contrast, the tutoring item is largely retrieval based and Socratic in format but not entirely agentic [57].

V. CHALLENGES AND LIMITATIONS

A. Socratic Scaffolding Quality

Socratic scaffolding was the least developed dimension of evaluation, with an average score of 3.00 out of 5. The system often generated quite superficial follow up questions rather than more probing inquiries that ask individuals to compare, reflect, or analyze outcomes. These findings suggest that, despite the tutor's ability to guide basic questioning, the Socratic prompting strategy is not robust across domains and query types. This finding aligns with earlier studies showing that maintaining sustained, high quality pedagogical guidance and deep interaction is difficult for AI tutoring systems [14].

B. Transparency and Source Attribution

Although the system achieved good context grounding performance, learners did not always refer to it in their interactions. Although retrieved passages were typically incorporated well into answers, the system was not always able to explicitly provide attribution or the source. The ongoing design dilemma for retrieval grounded educational systems is the trade off between conversational fluency and opaque source attribution. Too little attribution can undermine trust and traceability, while too much can interfere with the learning process [58].

Moreover, the evaluation was conducted on a limited set of 14 questions; therefore, the reported scores should be interpreted as preliminary evidence of system behavior rather than broad generalizable performance claims. Future work will expand the evaluation using a larger question set, student-facing studies, and baseline comparisons.

C. Computing and Data Resource

The system was tested and trained using two Tesla T4 chips with 16 GB VRAM and 4 bit quantized inference. Although it enabled cost effective, practical experimentation for deployment, the configuration limited the size of the model that could be inferred and the scale of experimentation. Although more advanced models, stronger rerankers, or pipeline based tutoring architectures could provide further improvements, these options were beyond the available hardware budget. The retriever fine tuning phase also used 929 query passage pairs, limiting the amount of supervised adaptation in retrieval. These trade offs align with the existing literature on model compression and the resource constrained implementation of a large transformer system [59].

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:58:37 UTC from IEEE Xplore. Restrictions apply.

VI. CONCLUSION AND FUTURE WORK

This paper presents an intelligent system based on a chatbot, intelligent Socratic tutoring and adaptive quiz generation system, retrieval augmented with large language models, for Arkansas Tech University. The system used a hybrid retrieval pipeline comprising FAISS, BM25, and cross encoder reranking of 2,328 chunks from 17 open educational resource textbooks, a retriever fine tuned on 929 domain specific query passages, and the Qwen3 8B language model. Assessment of 14 varied items yielded an overall score of 4.00/5, indicating that the system demonstrated strength in relevance, conciseness, contextual grounding, and hallucination resistance. Still, there are some areas for improvement, such as empathy and the quality of Socratic scaffolding. Further research will be conducted in four directions. The first is to enhance adaptive scaffolding, in which the tutor can respond dynamically by adjusting guidance based on learners' progress and recurring misconceptions, whilst maintaining a fruitful balance between Socratic prompting and direct support to prevent learner frustration [1], [7]. Second, the contribution of retrieval, reranking, and prompts, as independent variables, to the overall quality of tutoring requires controlled ablation studies [2]. Third, control over adaptive difficulty and alignment to the Taxonomy of Bloom should be added to the quiz generation module [60], [61]. Lastly, formal user tests assessing learning outcomes, cognitive load, and student satisfaction are needed to objectively evaluate the pedagogical efficacy of the system, beyond heuristic assessment [62].

REFERENCES

1 Joseph J. Slade, Alina Hyk, and Regan A. R. Gurung. Transforming learning: Assessing the efficacy of a retrieval-augmented generation system as a tutor for introductory psychology. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, volume 68, pages 1827–1830. SAGE Publications, 2024.

2 Yogesh Simmhan and Varad Kulkarni. Towards ai agents for course instruction in higher education: Early experiences from the field. *arXiv preprint arXiv:2510.20255*, 2025.

3 Erick S. Tyndall, Colleen Gayheart, Adama Some, Jordan Genz, Thomas Wagner, and Brian Langhals. Impact of retrieval-augmented generation and large language model complexity on undergraduate exams created and taken by ai agents. *Data & Policy*, 7:e57, 2025.

4 Yufeng Qian. Prompt engineering in education: A systematic review of approaches and educational applications. *Journal of Educational Computing Research*, 63(7-8):1782–1818, 2025.

5 Wei Qiu, Chit Lin Su, Nurabidah Binti Jamil, Maung Thway, Samuel Soo Hwee Ng, Lei Zhang, Fun Siong Lim, and Joel Weijia Lai. A systematic approach to evaluate the use of chatbots in educational contexts: Learning gains, engagements and perceptions. *Computers*, 14(7):270, 2025.

6 Jorge Fernández-Herrero. Evaluating recent advances in affective intelligent tutoring systems: A scoping review of educational impacts and future prospects. *Education Sciences*, 14(8):839, 2024.

7 Behnam Zakeri. Generative ai for project-based assessment in energy and sustainability education. *Research Square*, 2025. Preprint.

8 Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. *arXiv preprint arXiv:2005.11401*, 2020.

9 Jakub Swacha and Michał Gracel. Retrieval-augmented generation (rag) chatbots for education: A survey of applications. *Applied Sciences*, 15(8):4234, 2025.

10 Romain Puech, Jakub Macina, Julia Chatain, Mrinmaya Sachan, and Manu Kapur. Towards the pedagogical steering of large language models for tutoring: A case study with modeling productive failure. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 26291–26311, Vienna, Austria, 2025. Association for Computational Linguistics.

11 Jiayu Liu, Zhenya Huang, Tong Xiao, Jing Sha, Jinze Wu, Qi Liu, Shijin Wang, and Enhong Chen. Socraticlm: Exploring socratic personalized teaching with large language models. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024.

12 Nicy Scaria, Suma Dharani Chenna, and Deepak Subramani. Automated educational question generation at different bloom's skill levels using large language models: Strategies and evaluation. In *Artificial Intelligence in Education*, volume 14830 of *Lecture Notes in Computer Science*, pages 165–179. Springer, 2024.

13 Yuto Tomikawa, Ayaka Suzuki, and Masaki Uto. Adaptive question–answer generation with difficulty control using item response theory and pretrained transformer models. *IEEE Transactions on Learning Technologies*, 2024.

14 Moosa Ahmed Hassan Bait Ali Sulaiman. *Application of Teachers' Knowledge of Socratic Questioning in Developing EFL Critical Thinking Skills among Omani Post-Basic Learners*. Ph.d. dissertation, University Malaysia Sabah, 2019.

15 Hanyi Xu, Wensheng Gan, Zhenlian Qi, Jiayang Wu, and Philip S. Yu. Large language models for education: A survey. *arXiv preprint arXiv:2405.13001*, 2024.

16 Shen Wang, Tianlong Xu, Hang Li, Chaoli Zhang, Joleen Liang, Jiliang Tang, Philip S. Yu, and Qingsong Wen. Large language models for education: A survey and outlook. *arXiv preprint arXiv:2403.18105*, 2024.

17 Zhensheng Liu, Prateek Agrawal, Saurabh Singhal, Vishu Madaan, Mohit Kumar, and P. K. Verma. Lpitutor: an llm based personalized intelligent tutoring system using rag and prompt engineering. *PeerJ Computer Science*, 11:e2991, 2025.

18 Avijit Pandey, Gaurav Misra, Sunny Prajapati, and Syaamantak Das. Robotub: Retrieval-augmented tutoring for adaptive learning in stem moocs. In *Proceedings of the International Conference on Computers in Education (ICCE)*, dec 2025. [Online]. Available: https://library.apsce.net/index.php/ICCE/article/view/5938.

19 Eugenio Tufino. Notebooklm as a socratic physics tutor: Design and preliminary observations of a rag-based tool. *arXiv preprint arXiv:2504.09720*, 2025.

20 Negin Golchini, Elisabetta Passalacqua, Leah Vaughn, Raja-Elie E. Abdulnour, Theodore Zack, and Samuel Finlayson. Socratic ai: An adaptive tutor for clinical case based learning. *medRxiv*, 2025.

21 S. Gopi, D. Sreekanth, and N. Dehbozorgi. Enhancing engineering education through llm-driven adaptive quiz generation: A rag-based approach. In *2024 IEEE Frontiers in Education Conference (FIE)*, pages 1–8, Washington, DC, USA, 2024. IEEE.

22 Zeynep Karatepe, H. Esma Gözü, M. N. Alpdemir, and Y. Alpdemir. Using retrieval augmented generation (rag) and domain adaptation for high performance intelligent tutoring systems: A case study. In *2026 5th International Informatics and Software Engineering Conference (IISEC)*. IEEE, feb 2026.

23 Cesar Abraham Gomez-Cabello, Srinivasagam Prabha, Syed Ali Haider, Ariana Genovese, Bernardo G. Collaco, Nadia G. Wood, Sanjay Bagaria, and Antonio Jorge Forte. Comparative evaluation of advanced chunking for retrieval-augmented generation in large language models for clinical decision support. *Bioengineering*, 12(11):1194, 2025.

24 Ayman Asad Khan, Md Toufique Hasan, Kai Kristian Kemell, Jussi Rasku, and Pekka Abrahamsson. Developing retrieval augmented generation (rag) based llm systems from pdfs: An experience report. *arXiv preprint arXiv:2410.15944*, 2024.

25 Guanghui Qin, Yukun Feng, and Benjamin Van Durme. The nlp task effectiveness of long-range transformers. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 3774–3790, Dubrovnik, Croatia, 2023. Association for Computational Linguistics.

26 Raphaël Sourty, José G. Moreno, Lynda Tamine, and François Servant. Cherche: A new tool to rapidly implement pipelines in information retrieval. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*, pages 3283–3288, 2022.

27 Zahin Ibnat, Paul E. Calzada, Rasin Mohammed Ihtemam, Sujan Kumar Saha, Jingbo Zhou, Farimah Farahmandi, and Mark Tehranipoor. Deepv: A model-agnostic retrieval-augmented framework for verilog

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:58:37 UTC from IEEE Xplore. Restrictions apply.

code generation with a high-quality knowledge base. *arXiv preprint arXiv:2510.05327*, 2025.
28 Mashrur Rahman, Mantaqa Abedin, Monowar Zamil Abir, Md. Faizul Islam Ansari, Adib Reza, Farig Yousuf Sadeque, and Niloy Farhan. Transforming mentorship: An ai powered chatbot approach to university guidance. *arXiv preprint arXiv:2511.04172*, 2025.
29 Alberto Berenguer, Adriana Morejón, David Tomás, and José-Norberto Mazón. Leveraging large language models for sensor data retrieval. *Applied Sciences*, 14(6):2506, 2024.
30 Amal Abed, Ivan Lukic, Jörg K. H. Franke, and Frank Hutter. Increasing llm coding capabilities through diverse synthetic coding tasks. *arXiv preprint arXiv:2510.23208*, 2025.
31 Boxin Wang, Wei Ping, Peng Xu, Lawrence McAfee, Zihan Liu, Mohammad Shoeybi, Yi Dong, Oleksii Kuchaiev, Bo Li, Chaowei Xiao, Anima Anandkumar, and Bryan Catanzaro. Shall we pretrain autoregressive language models with retrieval? a comprehensive study. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7763–7786, Singapore, 2023. Association for Computational Linguistics.
32 Harshit Nainwani and Hediyeh Baban. Search is not retrieval: Decoupling semantic matching from contextual assembly in rag. *arXiv preprint arXiv:2511.04939*, 2025.
33 Muhammed Hasan Çelik and Xiaohui Xie. Efficient inference, training, and fine-tuning of protein language models. *iScience*, 28(10):113495, 2025.
34 Prabin R. Shakya, Ayush Khaneja, and Kavishwar B. Wagholikar. For clinical data extraction, qlora attains accuracy close to lora while requiring lower compute resources. *medRxiv*, 2025. Preprint.
35 Darrin Lea, James Ghawaly, Golden Richard III, Aisha Ali-Gombe, and Andrew Case. Rex86: A local large language model for assisting in x86 assembly reverse engineering. *arXiv preprint arXiv:2510.20975*, 2025.
36 Hudson K. Etkin, Kai J. Etkin, Ryan J. Carter, and Camarin E. Rolle. Differential effects of gpt-based tools on comprehension of standardized passages. *Frontiers in Education*, 10:1506752, 2025.
37 William B. Wood and Kimberly D. Tanner. The role of the lecturer as tutor: Doing what effective tutors do in a large lecture class. *CBE—Life Sciences Education*, 11(1):3–9, 2012.
38 Kathryn N. Thompson, Kimberley L. Chandler, Candice Morgan, Daniel Khashabi, Emily A. Delinski, and Benjamin Van Durme. Artificial intelligence as a co-tutor: Assessing the impact in the advanced learning virtual classroom. *Journal of Advanced Academics*, 36(4):714–745, 2025.
39 Jingru Lin, Chen Zhang, Stephen Y. Liu, and Haizhou Li. Ragcapbench: Benchmarking capabilities of llms in agentic retrieval augmented generation systems. *arXiv preprint arXiv:2510.13910*, 2025.
40 Shiyi Cao, Dacheng Li, Fangzhou Zhao, Shuo Yuan, Sumanth R. Hegde, Connor Chen, Charlie Ruan, Tyler Griggs, Shu Liu, Eric Tang, Richard Liaw, Philipp Moritz, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. Skyrl-agent: Efficient rl training for multi-turn llm agent. *arXiv preprint arXiv:2511.16108*, 2025.
41 Zhiyuan Li, Haisheng Yu, Guangchuan Guo, Nan Zhou, and Jiajun Zhang. Muisqa: Multi-intent retrieval-augmented generation for scientific question answering. *arXiv preprint arXiv:2511.16283*, 2025.
42 Hengran Zhang, Keping Bi, Jiafeng Guo, Jiaming Zhang, Shuaiqiang Wang, Dawei Yin, and Xueqi Cheng. Llm-specific utility: A new perspective for retrieval-augmented generation. *arXiv preprint arXiv:2510.11358*, 2025.
43 Xinming Wang, Jian Xu, Bin Yu, Sheng Lian, Hongzhu Yi, Yi Chen, Yingjian Zhu, Boran Wang, Hongming Yang, Han Hu, Xu-Yao Zhang, and Cheng-Lin Liu. Mr-align: Meta-reasoning informed factuality alignment for large reasoning models. *arXiv preprint arXiv:2510.24794*, 2025.
44 Andrew Ravenscroft. Promoting thinking and conceptual change with digital dialogue games. *Journal of Computer Assisted Learning*, 23(6):453–465, 2007.
45 Anastasios Ioannis Ntourmas, Yannis Dimitriadis, Sophia Daskalaki, and Nikolaos M. Avouris. Assessing learner facilitation in mooc forums: A mixed-methods evaluation study. *IEEE Transactions on Learning Technologies*, 15(2):265–278, 2022.
46 Nguyen-Thinh Le. How do technology-enhanced learning tools support critical thinking? *Frontiers in Education*, 4:126, 2019.
47 Michelene T. H. Chi, Stephanie A. Siler, Heisawn Jeong, Takashi Yamauchi, and Robert G. Hausmann. Learning from human tutoring. *Cognitive Science*, 25(4):471–533, 2001.
48 Arthur C. Graesser, Mark W. Conley, and Andrew Olney. Intelligent tutoring systems. In Kenneth R. Harris, Steve Graham, and Tim Urdan, editors, *APA Educational Psychology Handbook*, volume 3, pages 451–473. American Psychological Association, 2012.
49 Susanne P. Lajoie. Extending the scaffolding metaphor. *Instructional Science*, 33(5–6):541–557, 2005.
50 Benjamin D. Nye, Arthur C. Graesser, and Xiangen Hu. Autotutor and family: A review of 17 years of natural language tutoring. *International Journal of Artificial Intelligence in Education*, 24(4):427–469, 2014.
51 Leo Havemann, Simon Katan, Edward Anstead, Marco Gillies, Joanna Stroud, and Sarah Sherman. Digitally supported assessment. In Linda Amrane-Cooper, David Baume, Sally Brown, Stylianos Hatzipanagos, Patricia Powell, Sarah Sherman, and Alan Tait, editors, *Online and Distance Education for a Connected World*, pages 195–210. UCL Press, London, UK, 2023.
52 Indira Koneru. Exploring moodle functionality for managing open distance learning e-assessments. *Turkish Online Journal of Distance Education*, 18(4):129–141, 2017.
53 John Gardner, Michael O'Leary, and Li Yuan. Artificial intelligence in educational assessment: 'breakthrough? or buncombe and ballyhoo?'. *Journal of Computer Assisted Learning*, 37(5):1207–1216, 2021.
54 Nina B. Ødegaard, Hilde T. Myrhaug, Tone Dahl-Michelsen, and Yngve Røe. Digital learning designs in physiotherapy education: A systematic review and meta-analysis. *BMC Medical Education*, 21(1):48, 2021.
55 Phillip Dawson, Michael Henderson, Tracii Ryan, Paige Mahoney, Megan Phillips, Elizabeth Molloy, and David Boud. Technology and feedback design. In *Learning, Design, and Technology: An International Compendium of Theory, Research, Practice, and Policy*. Springer, 2023.
56 Ritu Lakhtakia, Farah Otaki, Laila Alsuwaidi, and Nabil Zary. Assessment as learning in medical education: Feasibility and perceived impact of student-generated formative assessments. *JMIR Medical Education*, 8(3):e35820, 2022.
57 Ahmed El-Hashash. Role of digital formative assessment in improving the assessment and monitoring of students' learning and their significance during the covid-19 pandemic. *Open Journal of Educational Research*, 2(1):9–12, 2022.
58 Melanie M. Cooper and Michael W. Klymkowsky. Let us not squander the affordances of llms for the sake of expedience: Using retrieval augmented generative ai chatbots to support and evaluate student reasoning. *Journal of Chemical Education*, 101(11):4847–4856, 2024.
59 Prakhar Ganesh, Yinfei Chen, Xinyun Lou, Muhammad Hassan Khan, Yinfei Yang, Dhruba Shao, Noah A. Smith Chen, Caiming Xiong, Richard Socher, and Luke Zettlemoyer. Compressing large-scale transformer-based models: A case study on bert. *Transactions of the Association for Computational Linguistics*, 9:1061–1080, 2021.
60 Sabina Elkins, Ekaterina Kochmar, Jackie C. K. Cheung, and Iulian Vlad Serban. How teachers can use large language models and bloom's taxonomy to create educational quizzes. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 23084–23091, 2024.
61 Anne Herrmann-Werner, Teresa Festl-Wietek, Friederike Holderried, Lea Herschbach, Jörg Griewatz, Ken Masters, Stephan Zipfel, and Moritz Mahling. Assessing chatgpt's mastery of bloom's taxonomy using psychosomatic medicine exam questions: Mixed-methods study. *Journal of Medical Internet Research*, 26(1):e52113, 2024.
62 Abdullah Alenezi and Abdulhameed Alenezi. Evaluating the effectiveness of chatbot-assisted learning in enhancing english conversational skills among secondary school students. *Education Sciences*, 15(9):1136, 2025.

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:58:37 UTC from IEEE Xplore. Restrictions apply.